

ACCESS HYDROLOGICAL DATASET

ERT 474/574

Open-Source Hydro Data Analytics

Sep 14th 2026

 **University at Buffalo** The State University of New York



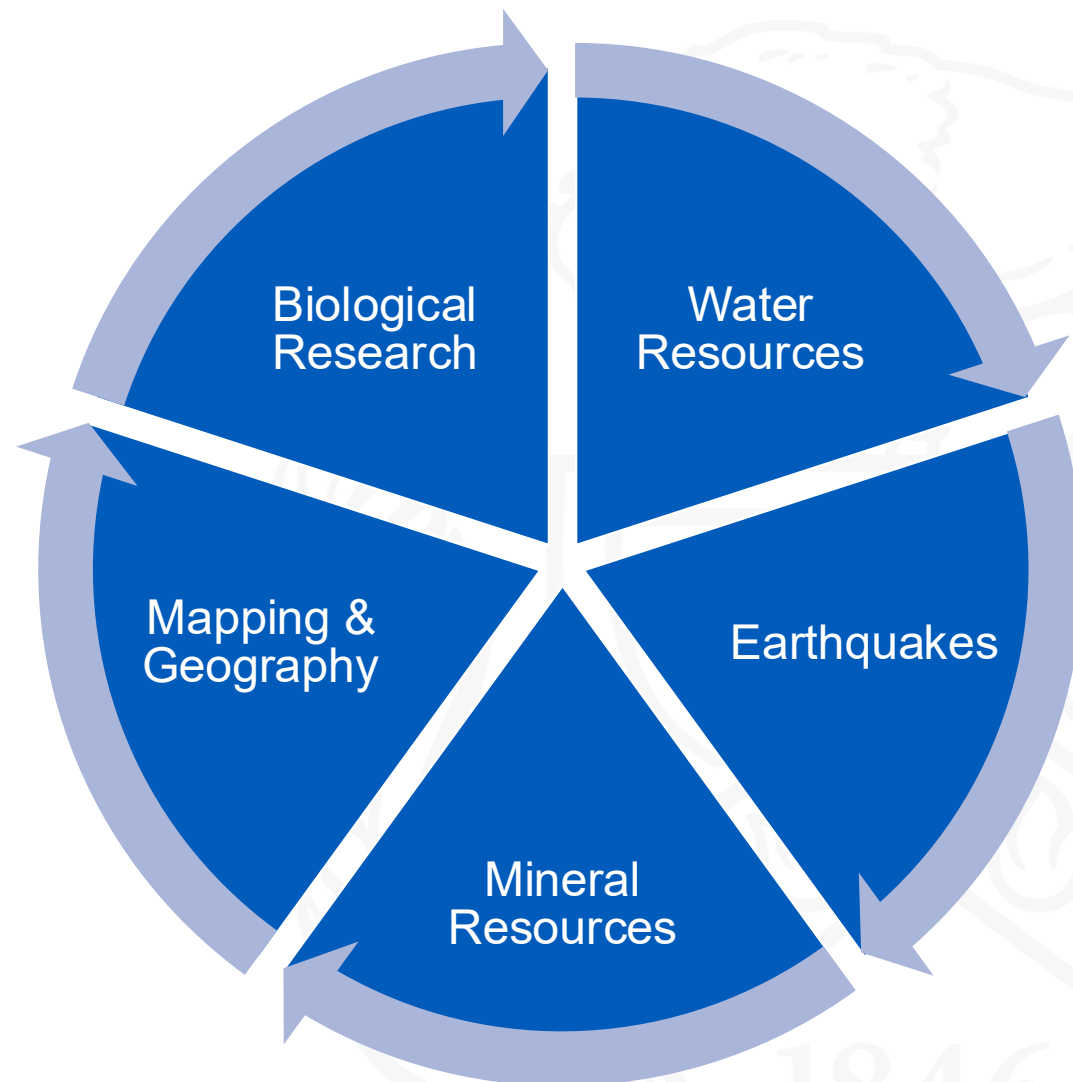
Logistics

- Any questions about homework #1?
 - How can we make sure that our homework is properly uploaded?
- Homework #2 – Numpy & Matplotlib practice
 - It is due 11:59 pm, Sep 21st (Monday)



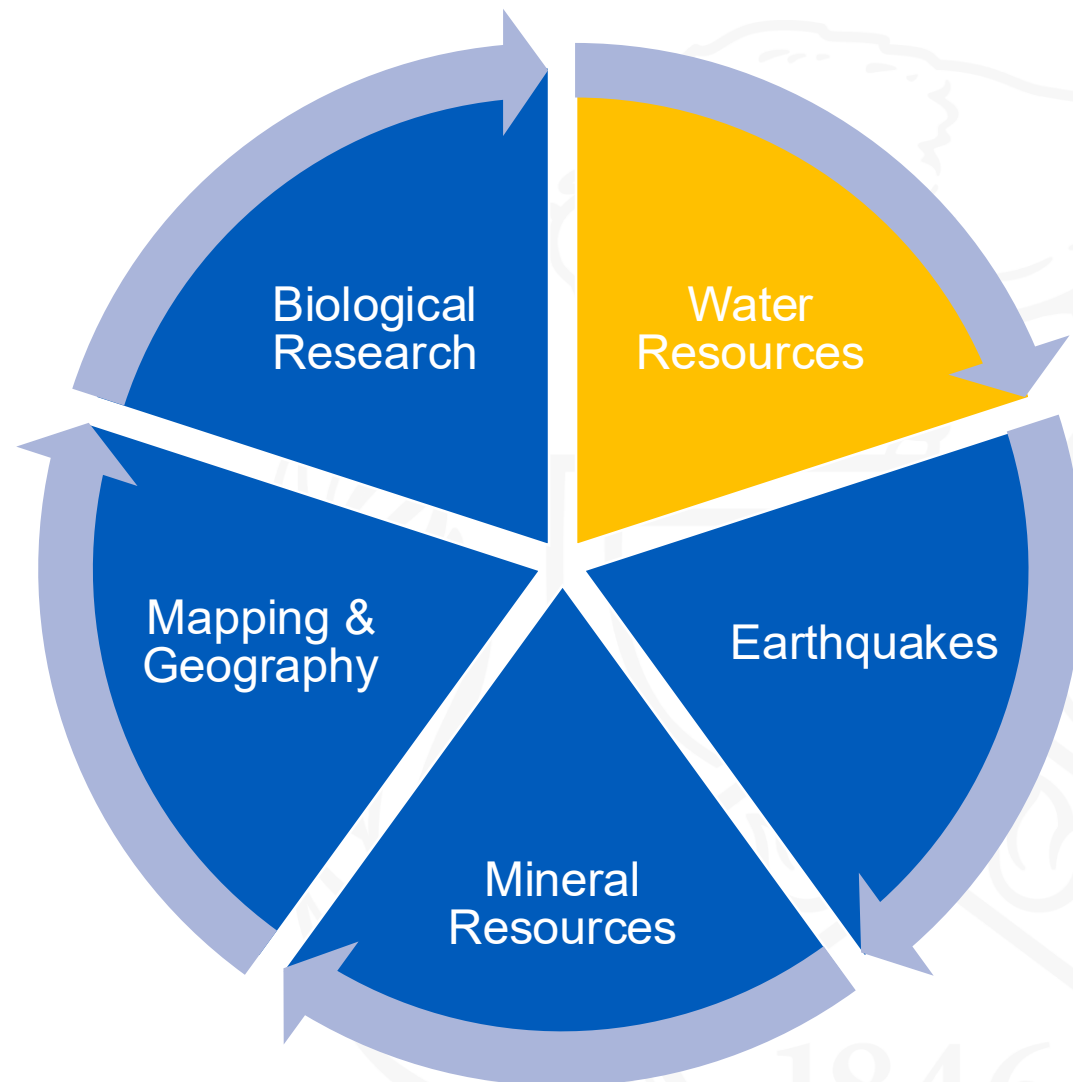
U.S. Geological Survey

- A science bureau within the United States Department of the Interior
- Mission: *to provide reliable, impartial information about the Earth's natural systems, enabling sound decisions regarding land use, resource management, and hazard mitigation.*



U.S. Geological Survey

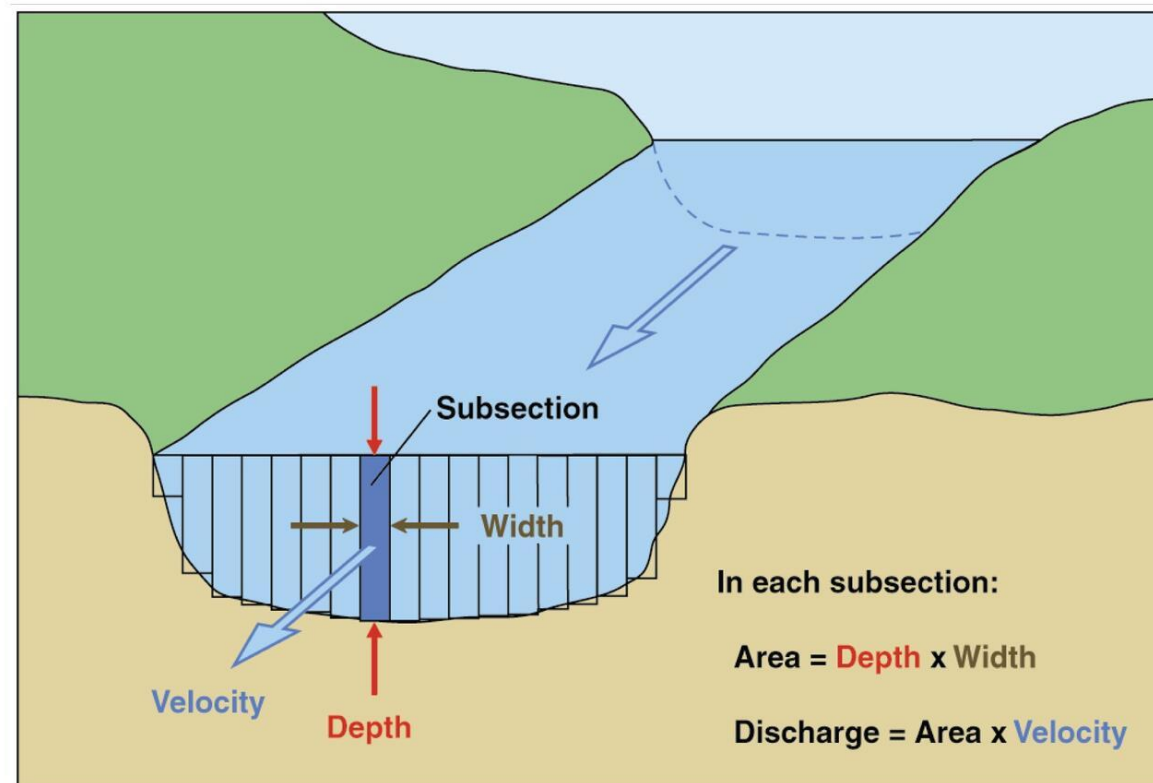
- A science bureau within the United States Department of the Interior
- Mission: *to provide reliable, impartial information about the Earth's natural systems, enabling sound decisions regarding land use, resource management, and hazard mitigation.*



How does USGS measure streamflow data?

Streamgaging

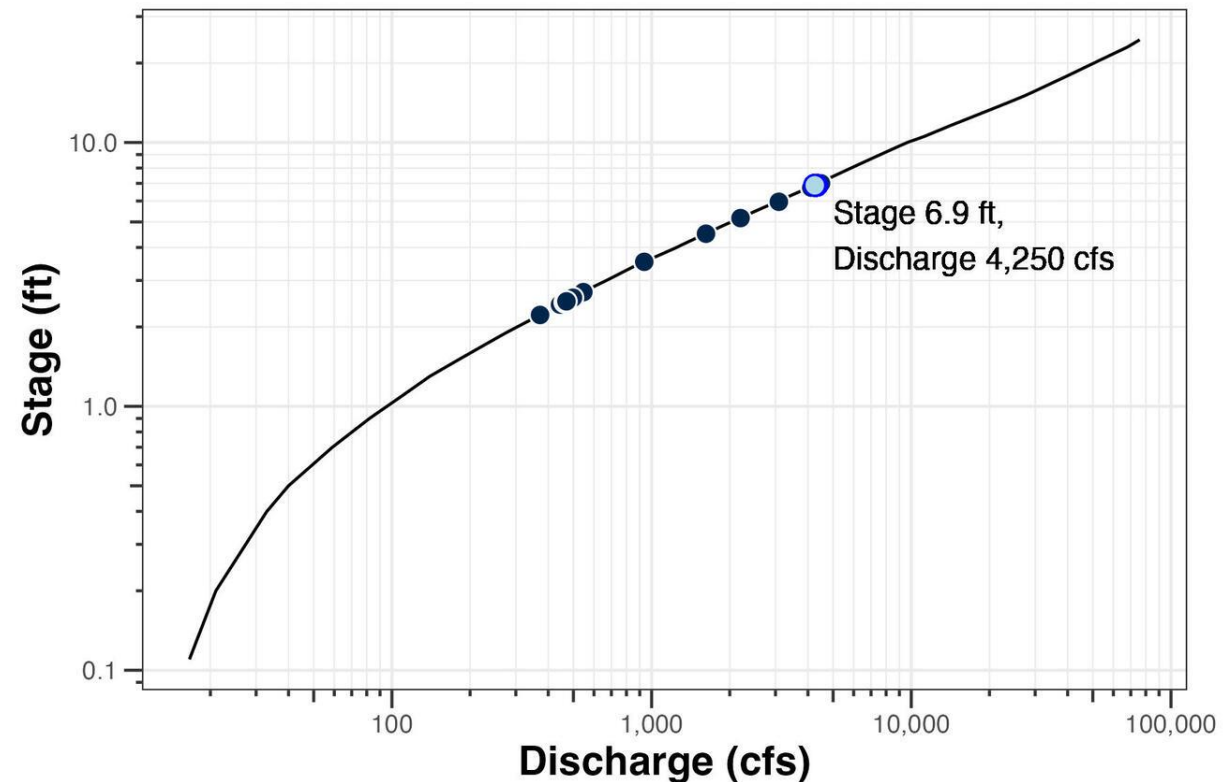
- The history of streamgaging can be dated back the late 19th century
- USGS started its first streamgaging in 1889 on the Rio Grande River in New Mexico to help determine if there was adequate water for irrigation purposes to encourage new development and western expansion.



How does USGS measure streamflow data?

Streamgaging generally involves 3 steps:

- **Measuring stream stage** (obtaining a continuous record of stage) - the height of the water surface at a location along a stream or river
- **The discharge measurement** (obtaining periodic measurements of discharge) - the quantity of water passing a location along a stream
- **The stage-discharge relation**—defining the natural but often changing relation between the stage and discharge; using the stage-discharge relation to convert the continuously measured stage into estimates of streamflow or discharge



Heads up: USGS moved everything

RETIRED — these no longer work

`waterdata.usgs.gov/nwis/sw`

- The legacy NWIS web pages have been retired
- Water data now lives on ***Water Data for the Nation***
- Machine access is now a standards-based web API
- Good news: the ideas are identical — we still build a URL and read what comes back

The screenshot shows the 'USGS Surface-Water Data for the Nation' legacy web interface. It features a header with the title and an 'Introduction' section. Below the header are four main data categories, each with a button and a site count: 'Current Conditions' (11,234 sites), 'Historical Observations' (19,248 sites), 'Daily Data' (29,803 sites), and 'Statistics' (27,794 sites). The 'Statistics' section has sub-buttons for 'Monthly', 'Annual', and 'Daily'. The 'Introduction' text describes the USGS National Water Information System (NWIS) and its data sources. The 'Daily Data' section provides a summary of the data available for each day.

USGS Surface-Water Data for the Nation

Introduction

The U.S. Geological Survey's (USGS) National Water Information System (NWIS) is a comprehensive and distributed application that supports the acquisition, processing, and long-term storage of water data. Water Data for the Nation serves as the publicly available portal to a geographically seamless set of much of the water data maintained within NWIS.

Nationally, USGS surface-water data includes more than 850,000 station years of time-series data that describe stream levels, streamflow (discharge), reservoir and lake levels, surface-water quality, and rainfall. The data are collected by automatic recorders and manual [field measurements](#) at installations across the Nation.

Data are collected by field personnel or relayed through telephones or satellites to offices where it is stored and processed. The data relayed through the Geostationary Operational Environmental Satellite (GOES) system are processed automatically in near real time, and in many cases, [current data](#) are available online within minutes.

Once a complete day of readings are received from a site, [daily summary data](#) are generated and made available online. USGS finalizes data at individual sites on a continuous basis as environmental conditions and hydrologic characteristics permit.

Current Conditions (11,234 sites)

Current conditions at selected sites based on the most recent data from on-site automated recording equipment. Measurements are commonly recorded at a fixed interval of 15- to 60-minutes and transmitted to the USGS every hour. Values may include "Approved" (quality-assured data that may be published) and/or more recent "Provisional" data (of unverified accuracy and subject to revision). Most current data are provisional.

Historical Observations (19,248 sites)

The same data accessed by the Current Conditions link above but including both active and discontinued sites with data for any part of the period October 1, 2007, through the present. Values may include "Approved" (quality-assured data that may be published) and/or more recent "Provisional" data (of unverified accuracy and subject to revision).

Daily Data (29,803 sites)

Summary of all data for each day for the period of record and may represent the daily mean, median, maximum, minimum, and/or other derived value. Values may include "Approved" (quality-assured data that may be published) and/or more recent "Provisional" data (of unverified accuracy and subject to revision).

Statistics (27,794 sites)

Monthly Annual Daily

This is the legacy interface!!

Water Data for the Nation

- The replacement for NWIS: real-time and historical water data
- <https://waterdata.usgs.gov>

How can we download streamflow data?!

USE THESE

Website (for exploration)

`waterdata.usgs.gov`

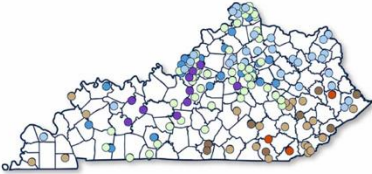
API (for downloading data)

`api.waterdata.usgs.gov/ogcapi/v0`

USGS GUI

- <https://waterdata.usgs.gov>
 - Go to explore page

Discover real-time data in your state



State pages provide a summary of the latest continuous data being collected in your state. Quickly filter the map and list to show a data type of interest to you, or customize the filters to further refine the map and list to your needs.

Select a State or Territory Page

Select a state

[Go to My Favorites](#)

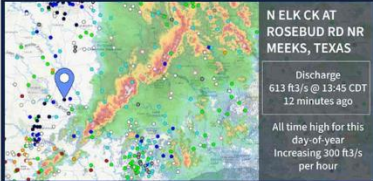
Create a list of your favorite monitoring locations



Make and share a list of USGS monitoring locations that you are interested in. Access the latest continuous water data at those locations all in one place. Create combined location graphs or a custom list of data graphs from your favorites list.

[Go to My Favorites](#)

View latest conditions and weather events



USGS has over 13,500 real-time stream, lake, reservoir, precipitation, water quality, and groundwater monitoring locations. View the latest continuous data in your area together with weather-related data to understand rapidly changing water conditions.

[Go to the National Water Dashboard](#)

Create a water alert



USGS monitors water conditions nationwide. Get notified with WaterAlert, a near-real time hydrologic alerting system that notifies you of stream, lake, and coastal conditions across the nation. Subscribe to text or email alerts when water conditions change.

[Go to WaterAlert](#)

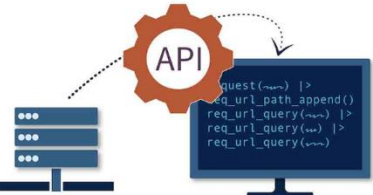
Explore historical water data



USGS has collected water data throughout US states and territories for more than 135 years. Find historical and currently operational monitoring locations collecting continuous and discrete sample data that interests you. Filter using locations of interest, type of water data, and recency of data that meet your needs.

[Go to Explore USGS Water Data](#)

Access water data



USGS has several options to support access to all water data available in Water Data for the Nation. Water Data APIs provide USGS water data in machine-readable formats via REST APIs, a common framework applications use to search and download data. The dataRetrieval package simplifies the process of loading USGS water data into R or Python environments.

[Go to Water Data APIs](#)

[Go to dataRetrieval](#)

USGS GUI

- <https://waterdata.usgs.gov>
 - Go to explore page

It will probably land on a random place!
Click the **show filter and sort options** to filter the sites!

The screenshot shows the USGS Water Data Explorer interface. The top navigation bar includes 'WDFN Home', 'WDFN tools and data', 'Other water data resources', and 'Connect'. A search bar is present with the text 'Search by map feature or location ID'. The main map area displays the Potomac River and surrounding regions, with several location pins. A yellow box highlights the 'Show filter and sort options' button. To the right of the map, a sidebar shows '11 locations found' and lists filters: 'Data collections: Continuous', 'Time range: 120 days', and 'Location Types: Surface water, Groundwater'. Below this, there are sections for 'My Favorites' (0 locations selected) and a list of locations. The first location is 'Groundwater' with 1 location found, and the second is 'Surface water' with 10 locations found. The 'Surface water' section is expanded, showing a list of locations with their USGS IDs and names.

Location	USGS ID
Broad Run Near Leesburg, VA	USGS-01644280
Dead Run at Whann Avenue Near Mclean, VA	USGS-01646305

USGS GUI

- <https://waterdata.usgs.gov>
- Go to explore page

Step 1 - Select data collection categories

Select data collection categories

- ☐ **Continuous data** are collected by automated sensors installed at a monitoring location.
[Learn about Continuous data](#)
- ☒ **Daily data** provide one data value to represent water conditions for the day.
[Learn about Daily data](#)
- ☐ **Field measurements** are physically measured values collected during a visit to the monitoring location.
[Learn about Field measurements](#)
- ☐ **Discrete sample data** are derived from manual field collection and laboratory analysis of physically collected samples collected during a visit to the monitoring location.
[Learn about Discrete sample data](#)

Step 2 - Choose time range for data

Show locations with new data in the last:

- ☐ 120 days
- ☐ 1 year
- ☐ 5 years
- ☐ 10 years
- ☐ 20 years
- ☒ All possible years

Step 3 - Set location type filter

Select one or more location types

- ☒ Surface Water
- ☐ Groundwater
- ☐ Spring
- ☐ Atmospheric
- ☐ Other

If we decide to look at the historical streamflow data set near Buffalo

Step 4 (optional) - Set parameter filters

Select up to 5 keywords

Discharge, cubic feet per second x

discharge

Discharge, cubic feet per second

Discharge, instantaneous, cubic feet per second

Bedload sediment discharge, daily mean, per unit width of river, composite samples, short tons per day per foot of width

Discharge, cubic meters per second

Discharge, instantaneous, cubic meters per second

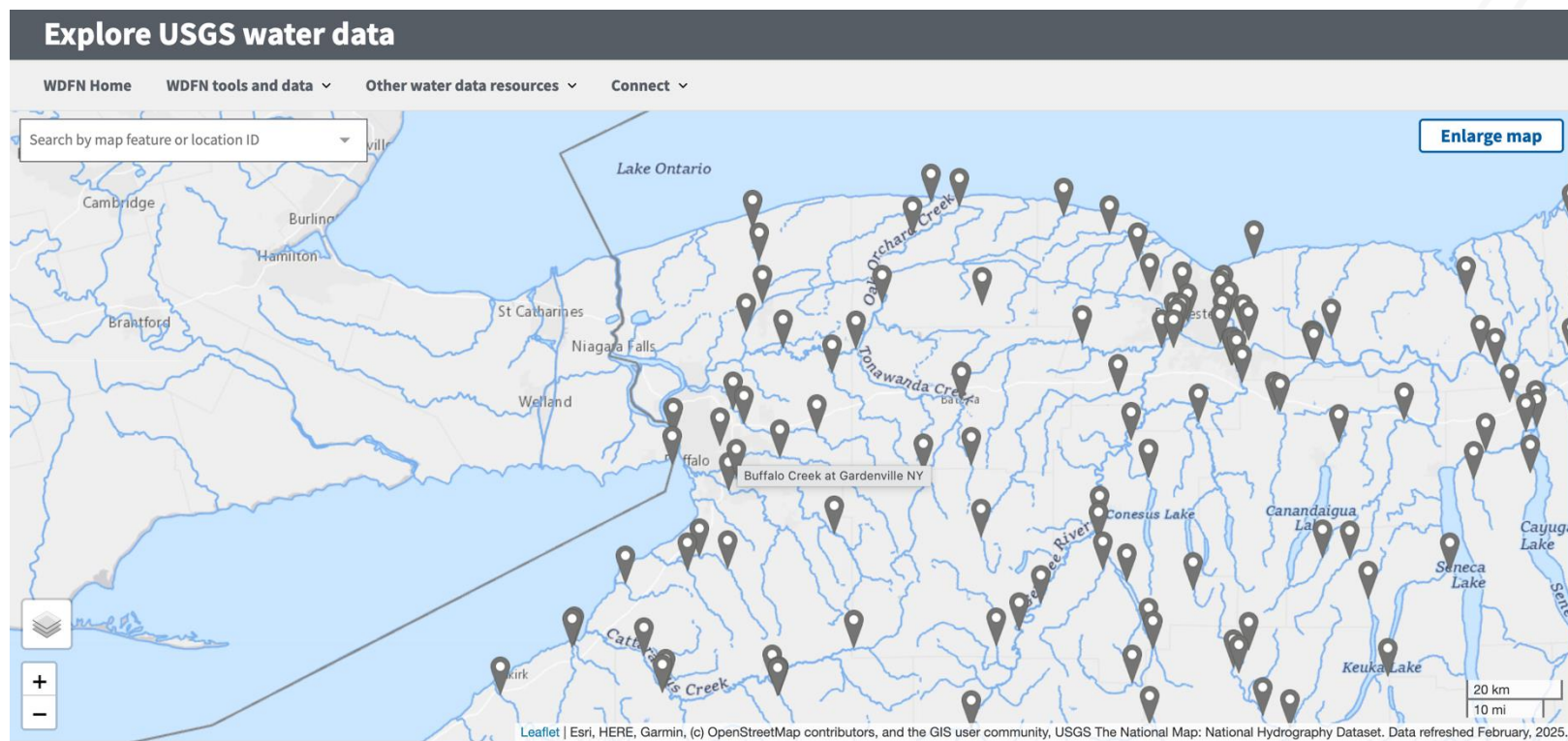
Discharge, gallons per minute

Discharge, peak for storm event, cubic feet per second

USGS GUI

- <https://waterdata.usgs.gov>
 - Go to explore page

Next, zoom left map to Buffalo



USGS GUI

- <https://waterdata.usgs.gov>
 - Go to explore page

We can randomly pick one site. There will be a link that leads you to a specific site page

Explore USGS water data

WDFN Home WDFN tools and data ▾ Other water data resources ▾ Connect ▾

i Important for you to know:

- How are we doing? We want to hear from you. Take our quick [survey](#) to tell us what you think.

Search by map feature or location ID ▾

2153 locations found

Using filters for:
Data collections: **Daily**
Time range: **All possible years**
Location Types: **Surface water**
Keywords: **Discharge, cubic feet per second**

Niagara River at Buffalo NY - USGS-04216000

Data collection: **Daily**
Location type: **Surface water**
Location subtype: **Stream**
Location ID: USGS-04216000
Keyword matches:
- **Discharge, cubic feet per second**
Parameter code matches: 00000

[View Monitoring location page](#)

☐ Select for My Favorites
[See My Favorites options](#)

Leaflet | Esri, HERE, Garmin, (c) OpenStreetMap contributors, and the GIS user community, USGS The National Map: National Hydrography Dataset. Data refreshed February, 2025.

Download USGS data

- Once you are done exploring the dataset, you may directly download the dataset by going to the following website.
- <https://api.waterdata.usgs.gov/download/>



Is there simpler ways for us to easily access it?

- Yes!
- More than one way too!!
 - We will introduce the first way today!
 - Access the USGS data using `urllib`



urllib

What's the url?

https://api.waterdata.usgs.gov/ogcapi/v1/collections/daily/items?monitoring_location_id=USGS-04216000¶meter_code=00060&statistic_id=00003&datetime=2022-10-01/2023-09-30&f=csv&limit=10000

**https://api.waterdata.usgs.gov/ogcapi/v1/collections/daily/items?
monitoring_location_id=USGS-04216000&
parameter_code=00060&
statistic_id=00003&
datetime=2022-10-01/2023-09-30&
f=csv&
limit=10000**

urllib

What's the url?

The base url

`https://api.waterdata.usgs.gov/ogcapi/v1/collections/daily/items?`

`monitoring_location_id=USGS-04216000&`

`parameter_code=00060&`

`statistic_id=00003&`

`datetime=2022-10-01/2023-09-30&`

`f=csv&`

`limit=10000`

urllib

What's the url?

“?” indicates the
beginning of a query

**https://api.waterdata.usgs.gov/ogcapi/v1/collections/daily/items?
monitoring_location_id=USGS-04216000&
parameter_code=00060&
statistic_id=00003&
datetime=2022-10-01/2023-09-30&
f=csv&
limit=10000**

urllib

What's the url?

Query parameters are
key-value pairs,
separated by the “=” sign

`https://api.waterdata.usgs.gov/ogcapi/v1/collections/daily/items?`

`monitoring_location_id=USGS-04216000&`

`parameter_code=00060&`

`statistic_id=00003&`

`datetime=2022-10-01/2023-09-30&`

`f=csv&`

`limit=10000`

urllib

What's the url?

Query parameters are
separated by the "&" sign

`https://api.waterdata.usgs.gov/ogcapi/v1/collections/daily/items?
monitoring_location_id=USGS-04216000&
parameter_code=00060&
statistic_id=00003&
datetime=2022-10-01/2023-09-30&
f=csv&
limit=10000`

urllib

What's the url?

station ID starts with
USGS

`https://api.waterdata.usgs.gov/ogcapi/v1/collections/daily/items?
monitoring_location_id=USGS-04216000&
parameter_code=00060&
statistic_id=00003&
datetime=2022-10-01/2023-09-30&
f=csv&
limit=10000`

urllib

What's the url?

00060 represents
discharge with unit *cfs*

`https://api.waterdata.usgs.gov/ogcapi/v1/collections/daily/items?
monitoring_location_id=USGS-04216000&
parameter_code=00060&
statistic_id=00003&
datetime=2022-10-01/2023-09-30&
f=csv&
limit=10000`

More parameter code can be accessed here
<https://api.waterdata.usgs.gov/ogcapi/v1/collections/parameter-codes/items>

Parameter codes

- <https://api.waterdata.usgs.gov/ogcapi/v1/collections/parameter-codes/items>

id	parameter_name	unit_of_measure	parameter_group_code	parameter_description	medium	statistical_basis	time_basis	epa_equivalent
<u>00060</u>	Discharge	ft ³ /s	PHY	Discharge, cubic feet per second	Water	Mean	1 Day	Not checked
<u>00010</u>	Temperature, water	deg C	PHY	Temperature, water, degrees Celsius	Water	None	None	Agree

urllib

What's the url?

00003 represents
mean values

Here, we download daily data,
so it refers to “daily mean”

`https://api.waterdata.usgs.gov/ogcapi/v1/collections/daily/items?
monitoring_location_id=USGS-04216000&
parameter_code=00060&
statistic_id=00003&
datetime=2022-10-01/2023-09-30&
f=csv&
limit=10000`

Statistic codes

- <https://api.waterdata.usgs.gov/ogcapi/v1/collections/statistic-codes/items>

id	statistic_name	statistic_description
00001	MAXIMUM	MAXIMUM VALUES
00002	MINIMUM	MINIMUM VALUES
00003	MEAN	MEAN VALUES
00004	AM	VALUES TAKEN BETWEEN 0001 AND 1200
00005	PM	VALUES TAKEN BETWEEN 1201 AND 2400
00006	SUM	SUMMATION VALUES
00007	MODE	MODAL VALUES
00008	MEDIAN	MEDIAN VALUES
00009	STD	STANDARD DEVIATION VALUES
00010	VARIANCE	VARIANCE VALUES
00011	INSTANTANEOUS	RANDOM INSTANTANEOUS VALUES
00012	EQUIVALENT MEAN	EQUIVALENT MEAN VALUES

urllib

What's the url?

Defining start date and end date, separated by "/"

```
https://api.waterdata.usgs.gov/ogcapi/v1/collections/daily/items?  
monitoring_location_id=USGS-04216000&  
parameter_code=00060&  
statistic_id=00003&  
datetime=2022-10-01/2023-09-30&  
f=csv&  
limit=10000
```

urllib

What's the url?

The format of this file is
csv. We also have “**json**”
option

**https://api.waterdata.usgs.gov/ogcapi/v1/collections/daily/items?
monitoring_location_id=USGS-04216000&
parameter_code=00060&
statistic_id=00003&
datetime=2022-10-01/2023-09-30&
f=csv&
limit=10000**

urllib

What's the url?

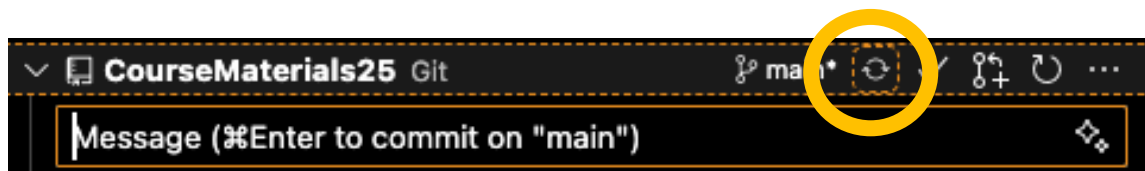
If we do not specify limit, it will only download 10 data points.

```
https://api.waterdata.usgs.gov/ogcapi/v1/collections/daily/items?
monitoring_location_id=USGS-04216000&
parameter_code=00060&
statistic_id=00003&
datetime=2022-10-01/2023-09-30&
f=csv&
limit=10000
```

This is a very rigid data format, so you might be thinking we can write some python code to automate this

- Practice

- Go to your own homework repo in github
- Use an existing codespaces
- Click  on the left panel
- Click  on "CourseMaterials26" repo



- The practice notebook will be available in `CourseMaterials26/coding_modules/lab3c_python_urllib.ipynb`

`https://api.waterdata.usgs.gov/ogcapi/v1/collections/daily/items?`

`monitoring_location_id=USGS-<SiteID>&`

`parameter_code=00060&`

`statistic_id=00003&`

`datetime=<Start-date>/<End-date>&`

`f=csv&`

`limit=10000`